

Theme 1 — What don't we know yet about AI, and what should we find out first?

This chapter helps an owner find the one AI gap that blocks a decision now.

[Team name] · [Cohort] · v0.6 · SME: persona — a fictional 40-person Dutch firm that designs and builds machines · Sources last appraised: 5 Oct 2026

The question, for this kind of SME

We are a 40-person firm that designs and builds machines in SOLIDWORKS, with PDM, ERP and spreadsheets. We record our own machine hours. **Which AI use do we test first, and how will we know it worked?**

The translation

- **What firms like ours do.** AI use at Dutch firms with 10–50 people rose to 27% in 2025, mostly for marketing and admin (CBS, 2025). Owners use Copilot or ChatGPT for quotes and keep a human in charge (Midpoint Brabant, 2025; van der Meer, 2025).
- **What they lack.** "Lack of experience" is the top barrier (73%) (CBS, 2025). 83% of SME manufacturers in Brabant do not know where AI fits in production (FME, 2026).

The firm says it needs	The evidence shows it needs	Difference
More experience (CBS, 2025)	A way to judge output: AI fails where users do not expect it (Dell'Acqua et al., 2023)	Experience alone does not show where AI fails
Legal clarity (CBS, 2025)	One duty already applies: AI literacy, since February 2025 (European Union, 2024)	The law is a reason to act, not to wait
(Not named)	Usable data: often "not in order" (Vankan et al., 2025); poor data makes models worse (OECD, 2025a)	The biggest difference: surveys do not ask about data (Eurostat, 2025)

The gap that matters most now

We do not know if our own data is good enough to test one AI use, so we cannot tell whether AI is right or pays off. Without a baseline, a payback rule such as "one year at most" (Midpoint Brabant, 2025) cannot be applied.

Illustrative use case (CRISP-DM, first steps)

1. Business understanding: write one goal and a payback limit (Linssen et al., 2025). **2. Data understanding:** check whether recorded hours link to a part or drawing in ERP and PDM. Theme 2 runs all six phases.

Recommended tooling

An Excel export from ERP and PDM, with Copilot or ChatGPT for formulas (van der Meer, 2025): a few days. A short in-house AI literacy session (European Union, 2024): a few hours.

Responsible AI & trust

- **Your data.** 52% of firms worry about data fed into AI (OECD, 2025b). Limit who may upload confidential files (van der Meer, 2025).
- **Rules.** Staff who use AI need AI literacy (European Union, 2024). Ask your adviser whether booked hours count as personal data under the GDPR.
- **People.** An expert accepts or rejects every AI suggestion (van der Meer, 2025). Name an owner and a back-up: a tool fell into disuse when its builder left (van Ginkel, 2025).

First five steps for an owner

1. **Name one decision and goal,** with a payback limit (Linssen et al., 2025; Midpoint Brabant, 2025).
2. **Measure your baseline:** compare quoted and actual hours on 20–30 recent jobs.
3. **Check your data:** can recorded hours be linked to parts in ERP or PDM? (FME, 2026; Vankan et al., 2025)
4. **Appoint an owner and a back-up,** and train AI users (European Union, 2024).
5. **Ask your ERP and SOLIDWORKS suppliers** what AI is coming (Midpoint Brabant, 2025). Then find out first: **is our data good enough to test AI cost estimation?** (Theme 2)

Theme 2 — What do firms like ours use AI for, and what pays off?

This chapter helps an owner test one AI use and tell whether it pays off.

[Team name] · [Cohort] · v0.4 · SME: persona — a fictional 40-person Dutch firm that designs and builds machines · Sources last appraised: 5 Oct 2026

The question, for this kind of SME

Theme 1 found that we do not know if our data is good enough to test AI. **Which AI use should we test first, and how will we know it works?**

The translation

AI use	Payoff evidence	Data needed
Chat assistant for reports and Excel	Measured: 2.8% of work hours saved on average (Humlum & Vestergaard, 2025)	Text, documents
Cost estimation for quotes	About 23.5% error with little data (Manuguerra et al., 2023); about 10% with 13,684 drawings (Arikan et al., 2025)	Drawings, material, batch size, actual hours (Jacobs, 2025)
AI-made drawings in SOLIDWORKS	No figures; output needs cleanup (Keglovits, 2025)	3D models, templates
Models on own data (custom GPTs, CAM)	Expected, not yet reached (van der Meer, 2025)	Own history, manuals

Only office assistants have measured gains, and they are small (Humlum & Vestergaard, 2025). **We chose cost estimation for quotes:** it runs on the data Theme 1 questions, own drawings and recorded hours (Jacobs, 2025), so testing it also tests the data gap. Start with in-house parts, not whole machines.

Illustrative use case: AI cost estimation (CRISP-DM)

Phase	What you do	Data reality
1. Business	Set a goal, pass level and payback limit (Linssen et al., 2025; Midpoint Brabant, 2025).	Build a baseline from past jobs.
2. Data	List models, drawings, quotes and hours.	Data use is weak at 80% of firms (FME, 2026).
3. Preparation	One table: past parts, quoted and actual hours.	Poor data makes models worse (OECD, 2025a).
4. Modelling	Trial a commercial platform first.	Gives a market average, not your costs (Jacobs, 2025).
5. Evaluation	Compare AI, human quote and actual cost.	AI can be wrong on unusual parts (Dell'Acqua et al., 2023).
6. Deployment	AI drafts; the estimator decides (van der Meer, 2025).	Without an owner, it falls into disuse (van Ginkel, 2025).

Pass level, set in advance: error no worse than your human quote (about 23.5% with little data (Manuguerra et al., 2023)), less time per quote, payback within one year (Midpoint Brabant, 2025).

Recommended tooling

Excel baseline: days. Quoting platform trial, such as The Quotation Factory (Brouwers, 2024): weeks. Own model with a knowledge centre (Linssen et al., 2025): months.

Responsible AI & trust

- **Your data.** Test on your own parts. Check client contracts before uploading their drawings.
- **Reliability.** Accuracy is tested only on other parts (Arikan et al., 2025; Manuguerra et al., 2023). Your own test in phase 5 is the real check.
- **People.** A named person approves every quote. Benefits rise 10–40% when employers encourage use (Humlum & Vestergaard, 2025).

First five steps for an owner

1. **Write the pass level down** before you start (Manuguerra et al., 2023; Midpoint Brabant, 2025).
2. **Build a test set** of 20–30 past in-house parts with quoted and actual hours and cost (Jacobs, 2025).
3. **Check contracts, then trial one quoting platform** on that set (Brouwers, 2024).
4. **Compare AI, human quote and actual cost** with your estimator (Dell'Acqua et al., 2023).
5. **Decide:** stop, use, or build your own model with a knowledge centre (Linssen et al., 2025).

Sources for Theme 1 and Theme 2

APA 7. Sources marked (translated from Dutch) are quoted in our English translation.

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